ELECTROCHEMISTRY

Introduction Electrochemistry is the branch of science which deals with the relationship between electrical energy and chemical energy and the inter-conversion of one form into other. If electrical energy brings about chemical reactions to occur, then the process is called electrolysis; for example, water can be made to split into hydrogen and oxygen by the passage of electric current. On the other hand, electrical energy can be produced as a result of chemical change. Both the above changes are called electrochemical changes and the latter finds use in the manufacture of cells and batteries, which generate electricity due to chemical changes. Both types of changes have widespread applications in chemistry. This chapter, however, deals only with the conversion of chemical energy into electrical energy and its applications in chemistry

16.3 Conductance in Electrolytic Solutions

When voltage is applied to electrodes dipped into an electrolytic solution and the ions of an electrolyte move, there is flow of electric current in the solution. This is termed as conductance which may be defined as the power of electrolytes to conduct electricity. Like metallic conductors, electrolytic solutions also obey Ohm's law.

Ohm's law The electric current flowing across a conductor is directly proportional to the potential difference across the conductor.

$$I \propto V$$

$$V = IR \text{ or } R = V/I \tag{1}$$

R = Proportionality constant called the resistance of the conductor in ohm (Ω)
I = Current in amperes
V = Potential difference in volts

Resistance It is the obstruction to the flow of current. It is expressed as R and its unit is ohm (Ω)

Specific resistance or resistivity: When a metallic conductor is a part of an electric circuit, the resistance R offered by it is directly proportional to the length l and inversely proportional to the area of cross section 'a' of the conductor. It can be expressed mathematically as

$$R \propto \frac{l}{a}$$

$$R = \rho \frac{l}{a} \tag{3}$$

where ρ(rho) is a constant called the 'specific resistance or resistivity' of the conductor. Its SI unit is ohm m (Ω m) although its sub-multiple ohm cm is frequently used. IUPAC recommends the use of the term resistivity over specific resistance. An electrolytic solution shows a similar behavior. In this case, the resistance offered by the solution is the resistance of the solution enclosed between the electrodes. l represents the length of the solution, that is, the distance between the electrodes and 'a' is the area of the smaller electrode (if they are of unequal size) exposed to the solution. In fact if we divide the volume of the solution enclosed between the electrodes by the distance between them we get the effective area of cross section of the volume of the solution.

If l = 1 and a = 1 then ρ = R or resistivity of a substance in a solution is defined as the resistance offered by a meter cube or centimeter cube of the solution.

1 ohm m = 100 ohm cm or 1 ohm cm = 0.01 ohm m

Conductance

It is the reciprocal of resistance and is a measure of the ease with which the current flows through a conductor. It is denoted by the symbol G

$$G = \frac{1}{R} \tag{3}$$

Its unit is ohm^{-1} (reciprocal ohm is also called mho represented by the symbol ℧). The SI units of conductance are siemens represented by the symbol S and 1S = 1ohm^{-1}.

Specific Conductance (Conductivity)

The reciprocal of resistivity (specific resistance) is called specific conductance or conductivity and is denoted by the Greek letter kappa (κ). As per the IUPAC recommendations, the term conductivity is used over specific conductance. It is measured in ohm^{-1}cm^{-1} or S cm^{-1}. The SI unit of conductance is seimens, length is m and area of cross section is m^2; hence the SI unit of conductivity (κ) is seimens per meter (S m^{-1}).

It may be noted that 1 S cm^{-1} = 100 S m^{-1}

$$\because \rho = R \times \frac{a}{l}$$

$$\frac{1}{\rho} = \frac{l}{R \times a} \quad \text{or} \quad \kappa = \frac{l}{R \times a} \tag{4}$$

Equivalent and Molar Conductivity

The conductivity of an electrolyte varies with concentration (it decreases with decrease in concentration); therefore, the conductivities of different electrolytes must be carried out at equivalent concentrations.

The equivalent conductivity of an electrolyte is the conductance of all ions produced by 1 g equivalent of an electrolyte. It is given by product of conductivity of the solution and the volume in cc containing 1 gram equivalent of the electrolyte. It is denoted by Λ (lambda). Thus if κ is the conductivity and V the volume in cubic centimeters containing 1 g equivalent of the electrolyte, then

$$\Lambda = \kappa \times V \tag{5}$$

If the concentration of the solution is c (i.e., c gm equivalent per liter), then the volume in liters which contains one gram equivalent of the electrolyte is 1000 / c.

Equation (5) can be written as

$$\Lambda = \kappa \times V = \frac{\kappa \times 1000}{c} \tag{6}$$

The units of equivalent conductivity are $ohm^{-1}\ cm^2\ equiv^{-1}$. Its SI unit is $S\ m^2\ equiv^{-1}$. Here, it is important to note that IUPAC has discontinued the use of the term equivalent conductivity and only molar conductivity is in use at present.

Molar conductivity (Λ_m): It is also used for the comparison of conductivities of different electrolytes. Molar conductivity is the conducting power of all ions produced by dissolving 1 g mole of an electrolyte in solution. It is given by the product of conductivity of the solution and the volume (V) in cc containing 1 gram mole of the electrolyte dissolved in it.

Mathematically,

$$\Lambda_m = \kappa \times V = \frac{\kappa \times 1000}{M}$$

where M is the concentration of the electrolyte in mole per liter. For a substance where equivalent and molecular weight are same its equivalent and molecular conductivities will have the same value. If we use $S\ cm^{-1}$ as the units for κ and $mol\ cm^{-3}$ as unit of concentration, then the units of molar conductivity will be $ohm^{-1}\ cm^2\ mol^{-1}$ or $S\ cm^2\ mol^{-1}$. However, if concentration is expressed in $mol\ m^{-3}$, then the unit of molar conductivity will be $S\ m^2\ mol^{-1}$, which is also its SI unit. Both the units are used and are interrelated as

$$1\ S\ m^2\ mol^{-1} = 10^4\ S\ cm^2\ mol^{-1} \text{ or}$$

$$1\ S\ cm^2\ mol^{-1} = 10^{-4}\ S\ m^2\ mol^{-1}$$

Measurement of conductivity The conductivity of an electrolytic solution can be obtained with the help of Eq. (4), if the resistance *R*, length *l* and area of cross-section *a* of the electrolytic conductor are known. Resistance is usually measured using a Wheatstone bridge. The solution whose conductivity is to be measured is placed in a special vessel called the conductivity cell. The cell consists of two electrodes made of platinum discs coated with finely divided platinum black and welded to a platinum wire fixed in two glass tubes. The glass tubes contain mercury and are tightly fixed in the ebonite cover of the cell so that the distance between the electrodes does not change during the experiment. Contact with the platinum discs is made by dipping the copper wires of the circuit in the mercury of the tubes (Fig. 16.1a). Since the use of direct current brings about electrolysis of the electrolyte and subsequent accumulation of the products at the electrodes, giving rise to back emf, alternating current is used and the galvanometer of the Wheatstone bridge is replaced by a headphone. The experimental set-up is shown in Figure 16.1b.

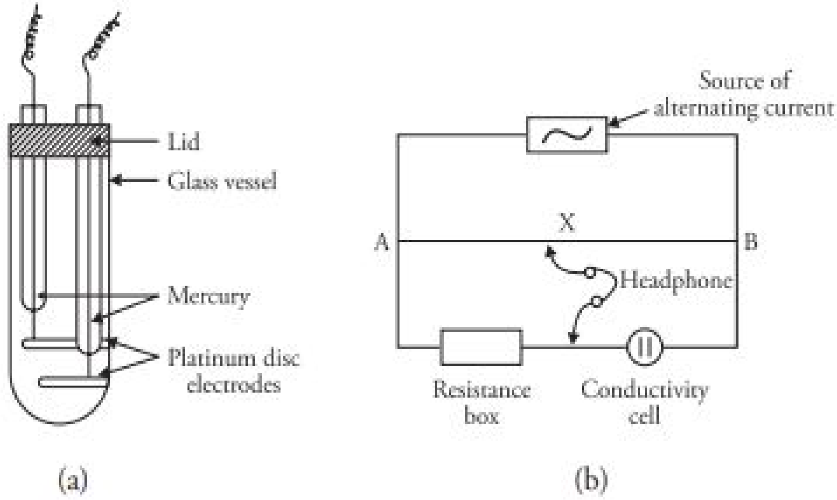

Figure 16.1 *(a) Conductivity cell*
(b) Experimental set-up for the measurement of conductivity of the solution of an electrolyte

AB is a wire of platinoid or manganin of uniform thickness stretched along the meter bridge. The bridge is connected to a source of alternating current. R is the resistance box in one arm of the bridge. The conductivity cell is placed in the second arm of the bridge. When the apparatus is connected, a sound will be heard in the headphone. At null point, no sound will be heard or the sound will be reduced to minimum. At this point, the bridge is said to be balanced at the null point.

$$\frac{\text{Resistance of the solution in the cell}}{\text{Resistance of resistance box (R)}} = \frac{BX}{AX}$$

$$\text{Resistance of the solution} = \frac{BX}{AX} \times R$$

Resistance of the solution can be calculated from the values of AX, BX and R.

Conductivity can be obtained by multiplying the resistance with the cell constant.

Cell constant

Cell constant is the ratio of distance between the two electrodes and the area of the electrode.

$$\text{Cell constant} = \frac{\text{Distance between the electrodes}}{\text{Area of the electrodes}}$$

From Eq. (4), we have

$$\text{Conductivity } \kappa = \frac{l}{R \times a}$$

It can be rewritten as $\kappa = \dfrac{\text{Cell constant}}{R}$

Cell constant = Conductivity × Resistance

or, Conductivity (κ) = Conductance × cell constant

or, Cell constant = $\dfrac{\text{Conductivity}}{\text{Conductance}}$ (Since Conductance = 1/R)

Solved examples

1. The conductivity (specific conductance) of 0.01 M KCl solution at 25 °C is 1.41×10^{-3} S cm^{-1}. The resistance of a cell containing this solution is 8.30 ohms. What is the cell constant of the cell?

   Solution

   Given R = 8.30 Ω $\kappa = 1.41 \times 10^{-3}$ S cm^{-1}.

   Cell constant = Conductivity (or specific conductance) × Resistance

   $= 1.41 \times 10^{-3} \times 8.30$

   $= 0.0117$ cm^{-1}

16.6 Conductometric Titration

Conductance measurements can be used to find the end point of reactions between electrolytes, precipitation reaction, etc. These measurements are particularly useful when the electrolytes are coloured and detection of the end point by the use of an indicator is difficult.

The principle involved in these titrations is based on the difference in conductance of various ions. Therefore, on gradual addition of a titrant to the titre, the conductance value changes. By plotting a graph between the volumes of titrant vs the conductance measurements, the end point of the reaction can be determined.

A brief description of different types of conductometric titrations is given below.

1. **Acid–base titrations**

   (i) **Titration of strong acid with a strong base**

   For such titrations, the strong acid (say HCl) is taken in the conical flask and its conductance is measured with the help of a conductivity cell. The conductance will be high as the acid contains H^+ ions which have maximum conductance. When a strong base (say NaOH) is added gradually from the burette the following ionic reaction takes place.

$$H^+ + Cl^- + Na^+ + OH^- \rightarrow Na^+ + Cl^- + \underset{\text{unionised}}{H_2O}$$

   From the above reaction, it can be seen that the fast moving H^+ ions are replaced by slow moving Na^+ ions (as the size of Na^+ is larger than H^+ their ionic conductance is less than the ionic conductance of H^+ ions); the conductance decreases on gradual addition of the alkali. After the neutralisation is complete, further addition of alkali results in the introduction of fast moving hydroxyl ions, increasing the conductance of the solution.

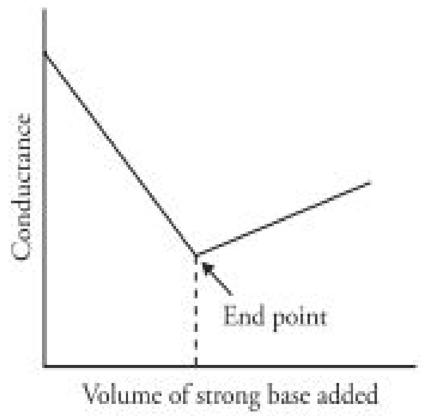

Figure 16.3 *Conductometric titration: strong acid and strong base*

   On plotting conductance against the volume of alkali added, a graph of the type shown in Figure 16.3 is obtained. It can be seen that the graph consists of two straight lines which intersect at the equivalence point.

   (ii) **Titration of strong acid and weak base**

   If a strong acid (such as HCl) is titrated with a weak base (like NH_4OH), the conductance falls initially due to the replacement of fast moving H^+ ions by slow moving NH_4^+ ions. However, after neutralisation, the conductance will remain almost constant, since the free base is feebly ionised as it is a weak electrolyte and will not contribute much to the conductance. Therefore, the second part of the curve is almost parallel to the *x*-axis.

$$H^+ + Cl^- + \underset{\text{Weak base}}{NH_4OH^-} \rightarrow [NH_4^+ + Cl^-] + H_2O$$

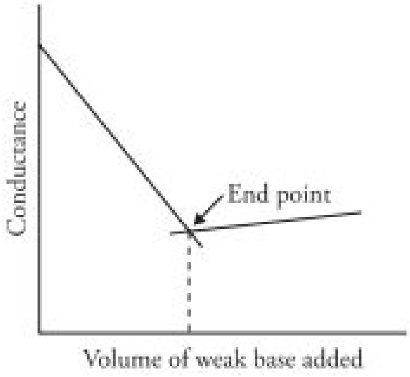

Figure 16.4 *Strong acid and weak base*

On plotting conductance against the volume of alkali added, a graph of the type shown in Figure 16.4 is obtained.

(iii) **Titration of weak acid and strong base**

When a weak acid like acetic acid is titrated against a strong base like sodium hydroxide, then the titration curve is obtained as shown in Figure 16.5. Although acetic acid is a weak acid, yet it furnishes some highly mobile H^+ ions which make a definite contribution to the conductance of the solution. On addition of base these H^+ ions are removed from the solution due to reaction with the base forming poorly ionised water, hence there is a slight decrease in conductance on addition of few drops of base initially. The initial decrease in conductance can also be attributed to the suppression of ionisation of acetic acid by the common acetate ions of strongly ionised sodium acetate. However, with further addition of base, the conductance of the highly ionised salt becomes greater than the conductance of the weak acid which it replaces and therefore, the conductance of the solution increases. After the neutralisation point, there is further increase in the conductance due to the presence of free base.

$$\underset{\text{Poorly conducting}}{CH_3COOH} + NaOH \rightarrow \underset{\substack{\text{Strongly ionised}\\ \text{(Highly conducting)}}}{CH_3COO^-Na^+} + \underset{\text{Un-ionised}}{H_2O}$$

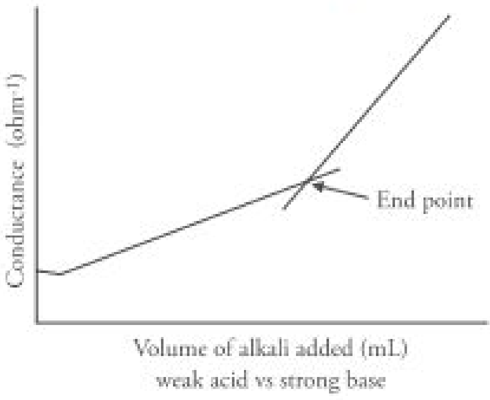

Figure 16.5 *Weak acid and strong base*

(iv) **Titration of a weak acid with a weak base**

Although acetic acid is a weak acid, it furnishes some highly mobile H^+ ions which make a definite contribution to the conductance of the solution. On addition of base these H^+ ions are removed from the solution due to reaction with the base ;moreover formation of ammonium acetate also suppresses the ionisation of acetic acid due to the common acetate ions (common ion effect). Both these factors lead to slight decrease in the conductance of the solution which can be seen in Figure 14.6.

$$\underset{\text{Weak acid}}{CH_3COOH} + \underset{\text{Weak base}}{NH_4OH} \rightarrow \underset{\text{ionised}}{NH_4^+ + CH_3COO^-} + \underset{\text{Un–ionised}}{H_2O}$$

Further addition of base increases the conductivity due to the formation of ionised salt ($CH_3COO^-NH_4^+$). After the neutralisation point, the conductance once again becomes constant due to free NH_4OH, which is a weak electrolyte and hence feebly ionised. On plotting conductance against the volume of alkali added, a graph of the type shown in Figure 16.6 is obtained.

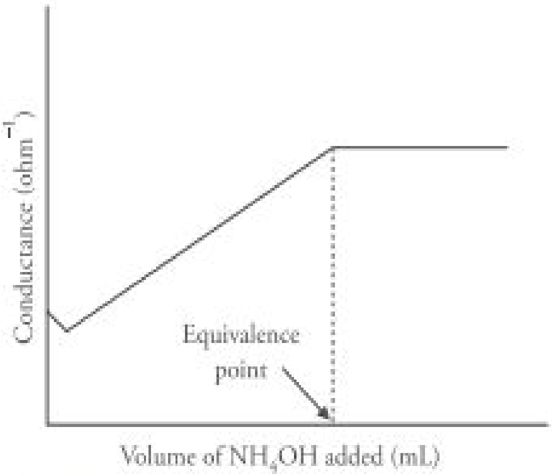

Figure 16.6 *Weak acid and weak base*

(v) **Precipitation titrations**

Precipitation titrations can also be followed conductometrically. Thus, when silver nitrate is added to a solution of potassium chloride, silver chloride precipitate is formed.

$$[K^+ + Cl^-] + [Ag^+ + NO_3^-] \rightarrow \underset{\text{Almost un-ionised}}{AgCl\downarrow} \quad + \quad [K^+ + NO_3^-]$$

In the above reaction, AgCl is precipitated out and K^+ ions remain unchanged. The only change that occurs is the replacement of Cl^- ions by NO_3^- ions. Since the mobility and molar ionic conductivity of these ions is almost the same (λ^-_{Cl} = 76.3 and $\lambda^-_{NO_3}$ = 71.4),

there is no change in conductivity on addition of $AgNO_3$. Addition of $AgNO_3$ after the equivalence point brings about a sharp increase in conductance due the addition of free Ag^+ and NO_3^- ions. On plotting conductance against the volume of $AgNO_3$ added, a graph of the type shown in Figure 16.7 is obtained.

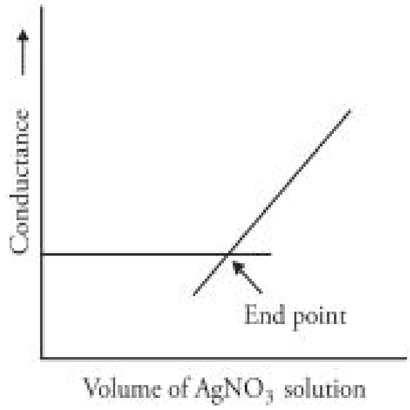

Figure 16.7 *Titration of KCl solution against $AgNO_3$*

2. **Advantages of conductometric titrations**

   Conductometric titrations have the following advantages.

   (i) They are useful for the analysis of dilute solutions.

   (ii) They can be used for the titration of a mixture of weak acid and weak base, which do not otherwise give sharp colour changes at end points.

   (iii) Mixtures containing coloured ions can be titrated.

   (iv) Since the end point is obtained graphically, errors due to visual observation of end point (colour change) are eliminated. Hence, fairly accurate results are obtained.

Glass electrode It is the most useful and versatile of all electrodes used for pH measurements. It is based on the principle that when two solutions of different hydrogen ion concentrations are separated by a glass membrane, a potential difference develops across the glass membrane. The magnitude of the potential difference depends upon the difference in the concentration of the hydrogen ions in the two solutions. The glass membrane is a special type of membrane having the approximate composition of 72% SiO_2, 22% Na_2O and 6% CaO.

A glass electrode consists of a thin walled bulb of pH sensitive glass sealed to a stem of non-pH sensitive high resistance glass (Fig. 16.13). Inside the glass electrode is a silver-silver chloride electrode or calomel electrode in 0.1 M HCl solution.

This glass electrode is dipped in a solution of varying hydrogen ion concentration (unknown pH) and is connected to an external reference electrode, usually a calomel electrode, with the help of a salt bridge.

The cell may be represented as

Ag, AgCl, 0.1 M HCl / Glass membrane / Experimental solution / KCl, Hg_2Cl_2(s), Hg

To find pH, this is coupled with a calomel electrode (Fig 16.18). The cell is formulated as
Ag / AgCl (s) / HCl (0.1 M)/ glass // solution of unknown pH/ calomel electrode

$$pH = \frac{E_{cell} - E_{calomel} + E^0_{glass}}{0.0591} \text{ at } 25\ ^\circ C$$

- E_{cell} = cell potential of the cell
- E^0 = cell potential under standard conditions

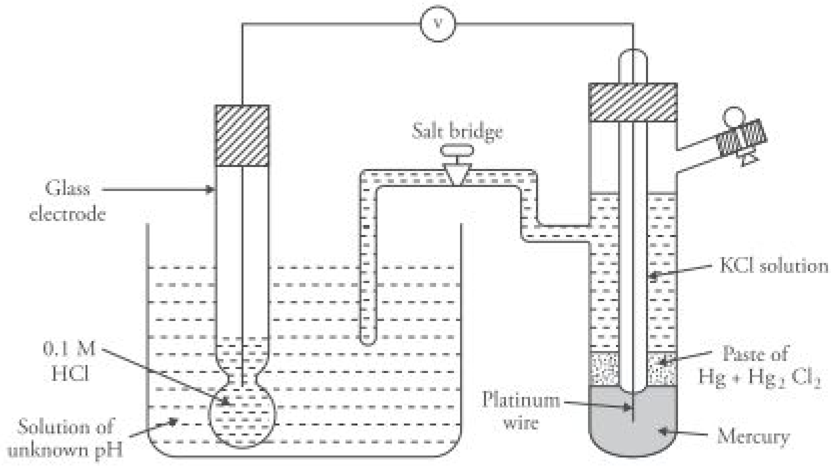

Figure 16.18 *Determinatin of pH by glass electrode*

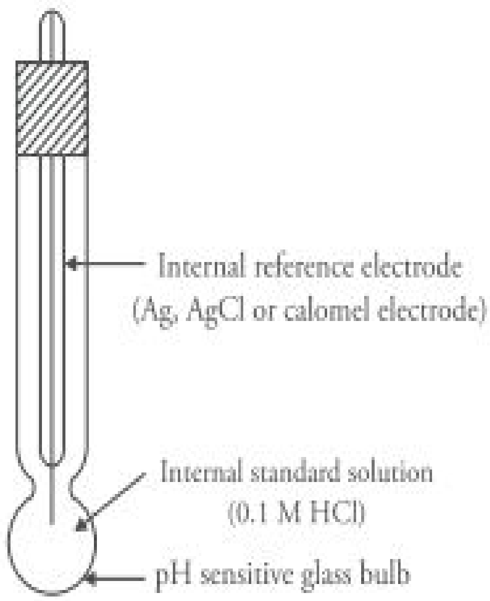

Figure 16.13 *Glass electrode*

16.18 Fuel Cells

Fuel cells are energy converters which take chemical substances from the external source and convert them to electrical energy. Fuel cell differs from a battery in that the chemical substances are an integral part of a storage battery whereas in fuel cells, they are fed into the cells whenever energy is desired. The electrodes used in fuel cells are generally inert but have catalytic properties. The anodic materials used are generally gaseous or liquids (contrary to metal anodes used in batteries).

In a fuel cell, hydrogen enters at the anode where it loses electrons. These electrons travel to the cathode. Oxygen enters at the cathode. It picks up electrons and either

- Combines with the hydrogen ions that travel from the anode through the electrolyte or
- Travels to the anode through the electrolyte and combines with hydrogen ions.

The hydrogen and oxygen ions combine to form water which is removed from the cell. A fuel cell generates electricity as long as hydrogen and oxygen are supplied to it.

The electrolyte plays a vital role in a fuel cell. It permits appropriate ions to pass between the anode and cathode. Depending on the type of electrolyte used, there can be different types of fuel cells. They are listed below.

- Hydrogen–oxygen fuel cells
- Phosphoric acid fuel cells
- Molten carbonate fuel cells
- Solid polymer electrolyte fuel cells
- Solid oxide fuel cells

Liquid electrolyte is used in the first three fuel cells whereas the last two use solid electrolyte.

Some cells use pure hydrogen, hence a 'reformer' is employed to purify the fuel. Some fuel cells can accept impurities but their efficient working requires a high temperature. Let us discuss these five types of fuel cells one by one.

Hydrogen–oxygen fuel cell

Hydrogen–oxygen fuel cell consists of porous screens of graphite coated with a layer of platinum catalyst (Fig. 16.25). The technology employed is proton exchange membrane fuel cell (PEMFC) technology. The electrolyte is a polymeric ion exchange membrane (polystyrenesulphonic acid or polymeric forms of perfluorosulphonic acid can be used). Moisture is to be supplied to keep the membrane wet hence the water balance is maintained in the resin by means of a wick. The water formed during the cell reaction is drained out and used for drinking. Several such cells are connected together to obtain the desired voltage.

Electrode reactions

Anodic reaction $H_2 \rightarrow 2H^+ + 2e^-$

Cathodic reaction $\frac{1}{2} O_2 + 2H^+ + 2e^- \rightarrow H_2O$

Overall cell reaction $H_2 + \frac{1}{2} O_2 \rightarrow H_2O$

The EMF of the cell at 25 °C is about 1.2 V. The above cell uses solid electrolyte and hence is called a solid polymer electrolyte cell. These cells are used in spacecrafts.

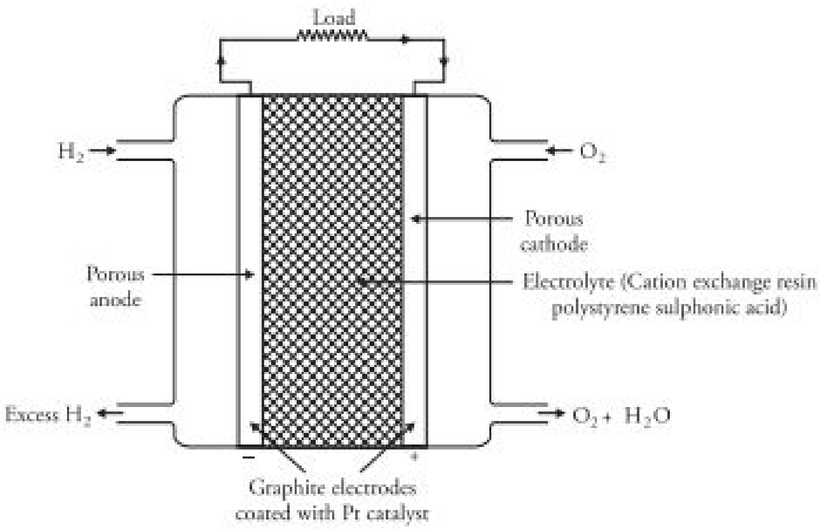

Figure 16.25 *$H_2 - O_2$ fuel cell (with solid polymer electrolyte)*

In another form of $H_2 - O_2$ fuel cell, alkali (aqueous 25% KOH at 200 °C) is used as an electrolyte (Fig. 16.26) and the electrodes consists of two inert porous electrodes made up of either graphite impregnated with finely divided platinum or a 75/25 alloy of palladium (Pd) with silver (Ag) or nickel (Ni). Hydrogen and oxygen are bubbled through the anode and cathode respectively. The cell reactions are

Anodic reaction $H_2 + 2OH^- \rightarrow 2H_2O + 2e^-$

Cathodic reaction $\frac{1}{2} O_2 + H_2O + 2e^- \rightarrow 2OH^-$

Net cell reaction $H_2 + \frac{1}{2} O_2 \rightarrow H_2O$

The EMF of the cell is nearly 1.23 volts.

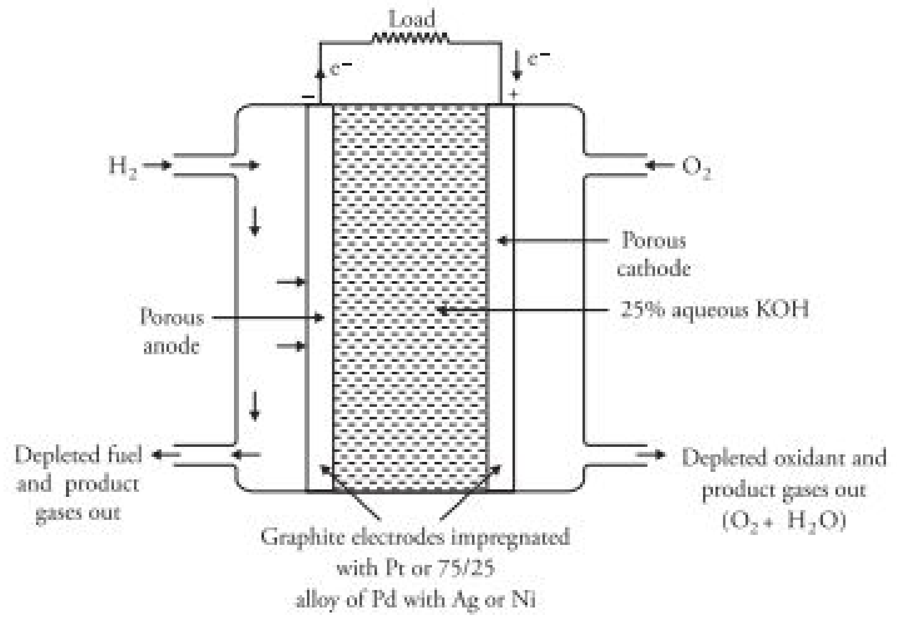

Figure 16.26 *H_2–O_2 fuel cell (with alkaline electrolyte)*